



## **NSBM Green University**

### **Faculty of Computing**

### **BSc(honors) in Software Engineering (UGC) –25.1**

### **IS201.3-Business Processes and ERP**

### **LUMINA UNIVERSITY ERP**

### **Business Process, ERP Design & Deployment Report**

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## 1. Business Process Analysis

Lumina is a comprehensive University Enterprise Resource Planning (ERP) system designed to streamline higher education operations. The academic institution runs on a continuous, standard loop: admit students, assign faculty, register courses, track daily attendance, conduct examinations, and issue final results. Historically, universities have relied on a patchwork of spreadsheets, standalone admissions software, and paper-based attendance registers, each maintained by a different department. This fragmentation makes it difficult to get a single, trustworthy view of any student or course at a given point in time.

Underneath this operational loop sit two core lifecycles that every academic ERP must handle: Student Lifecycle Management and Academic Administration. Previously, these processes operated in silos or on paper, meaning the same piece of information — a student's name, a lecturer's department, a course code — could be typed in three or four different systems, each with its own risk of typos, delays, and version conflicts.

### 1.1 Core Lifecycles

1. Student Lifecycle Management: Student admission → Course enrollment → Academic tracking → Graduation. This lifecycle follows an individual student from the moment they are accepted into the university through to the moment they receive their final degree classification.
2. Academic Administration: Faculty onboarding → Course assignment → Attendance tracking → Examination grading. This lifecycle follows the institution's teaching staff, from the point they join a department through to their day-to-day delivery of classes and assessment of student work.

Both flows exist to ensure the university functions, but in a disconnected system they require redundant data entry — the same admission, enrollment, or grading event often has to be typed into more than one place. The Lumina ERP changes how each step is recorded, ensuring that every action updates a single, shared academic record instead of isolated files. This removes duplicate work for administrative staff, reduces the chance of data entry errors, and gives every stakeholder — students, lecturers, and administrators — access to consistent, up-to-date information at all times.

### 1.2 Mapping to Standard ERP Modules

Each business process maps directly onto a standard ERP category, providing a clear framework for how the university operates at an enterprise level. This mapping is important because it shows that Lumina is not a bespoke, one-off system; it follows recognised ERP design patterns that are used across industries, adapted specifically for the higher-education context.

| Business Process                | Our Module             | Standard ERP Area                      |
|---------------------------------|------------------------|----------------------------------------|
| Admitting & tracking students   | Student Management     | Student Information System (SIS) / CRM |
| Faculty & staff administration  | Employee Management    | Human Capital Management (HCM)         |
| Curriculum & subject allocation | Course Registration    | Academic Operations / Core ERP         |
| Assessments & grading           | Examination Management | Evaluation Management                  |
| Tracking student presence       | Attendance Management  | Time & Tracking Management             |
| System metrics & analytics      | Dashboard & Overview   | Business Intelligence (BI)             |